

Reward Function Design

$$r = r_{\text{prod}} + r_{\text{safety}} + r_{\text{energy}} + r_{\text{stab}}$$

$$r_{\text{prod}} = q_{\text{CH}_4}/2000$$

$\approx 0.75 \sim 1.25$ (dominant positive term)

r_{safety}

General form ($x_i \in \{\text{VFA}, \text{pH}, \text{NH}_3\}$):

$= 0$ if x_i in safe range

$= -(c_i + w_i \cdot \varepsilon_i)$ otherwise

ε_i = deviation from boundary

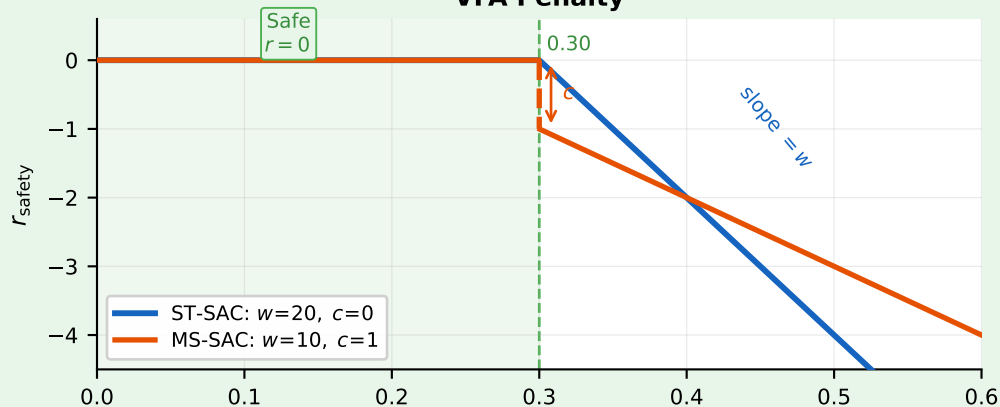
$$r_{\text{energy}} = -(q_{\text{ad}}/300)^2 \times 0.2$$

pump power cost

$$r_{\text{stab}} = -|\Delta q_{\text{CH}_4}|/100 \times 0.1$$

CH_4 volatility penalty

VFA Penalty



pH Penalty

